Can Your Clusters Do More?

Getting the most from your clusters with *Lightweight Distributed Metric Service* (LDMS)

NexGen Analytics (NGA) - November 5, 2026

Summary

NGA - *Lightweight Distributed Metric Service* (LDMS)

1. Background and Objectives
2. Current Observations
3. Method
4. Preliminary Results
5. Next Steps

Background and Objectives

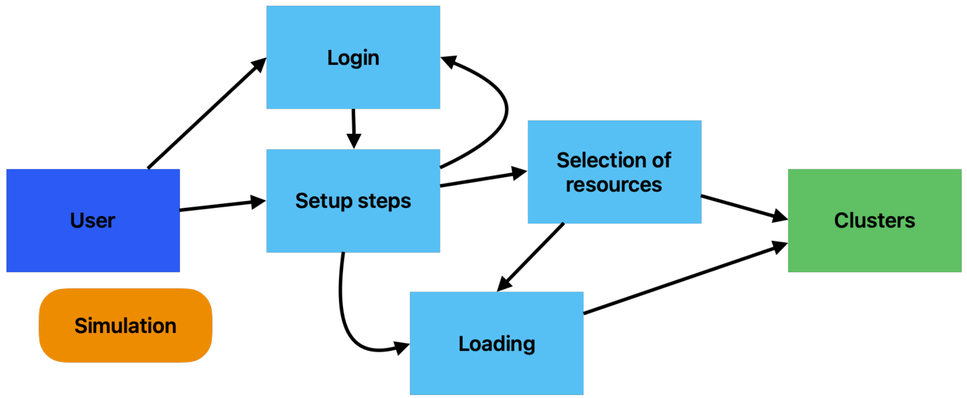
NGA - *Lightweight Distributed Metric Service* (LDMS)

- Ecosystem: *Open-source Varnish Information System* (OVIS)
- Work subsystem: *Lightweight Distributed Metric Service* (LDMS)
- Goals:
    1. Offload the user's workload to the system
    2. Predict the execution load of a job
        A. *Central Processing Unit* (CPU) usage
        B. Memory usage
    3. Save energy through better cluster management

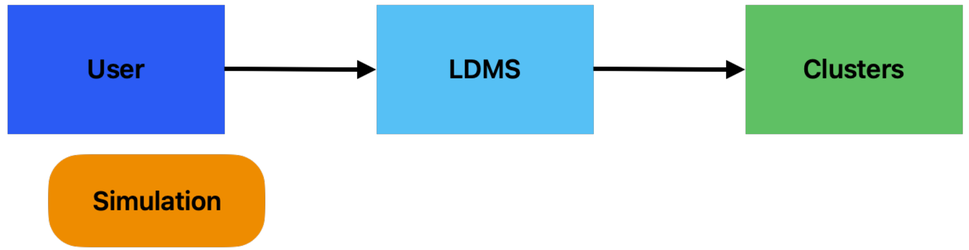
Current Observations 1

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- Users face:
  - Waiting lists
  - Login forms
  - Competition for resources



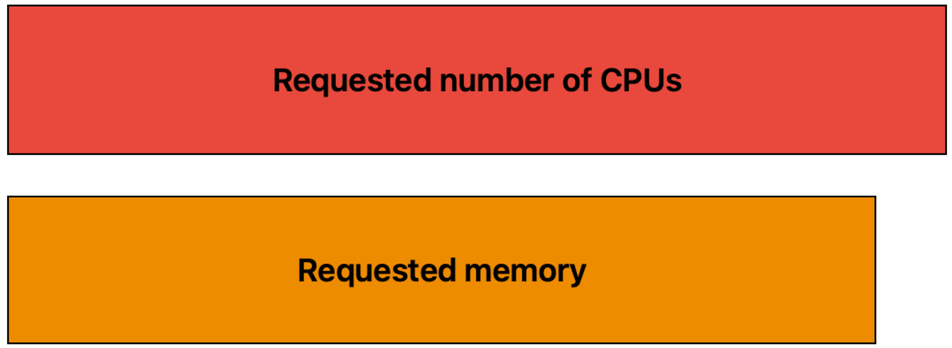
- Users want:
  - Have a simulation
  - Want to run it
  - Want to get the results



Current Observations 2

NGA - *Lightweight Distributed Metric Service* (LDMS)

- Users must:
  - Select the requested resources
  - Allow for a margin
  - Determine this margin
- Users want:
  - Not to worry about resources
  - Fast results

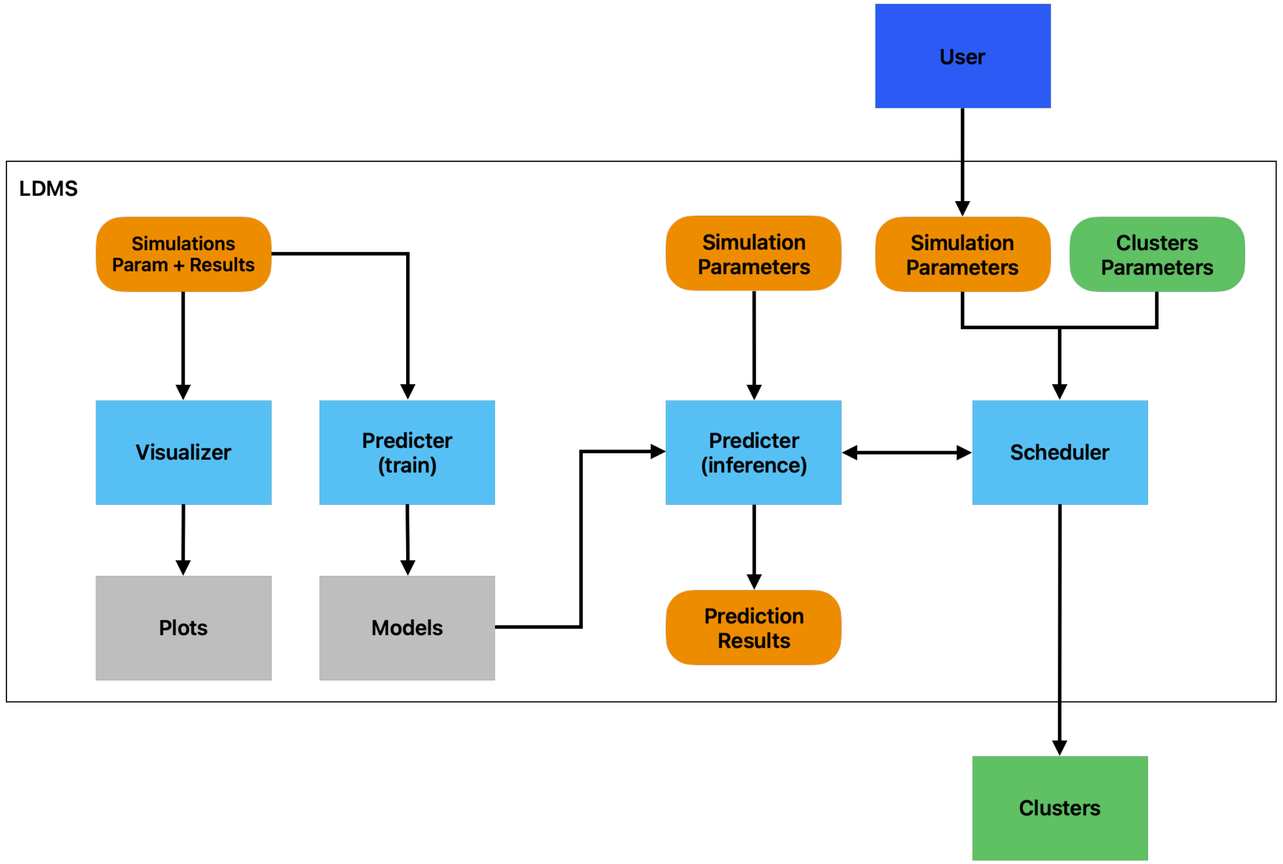




Method

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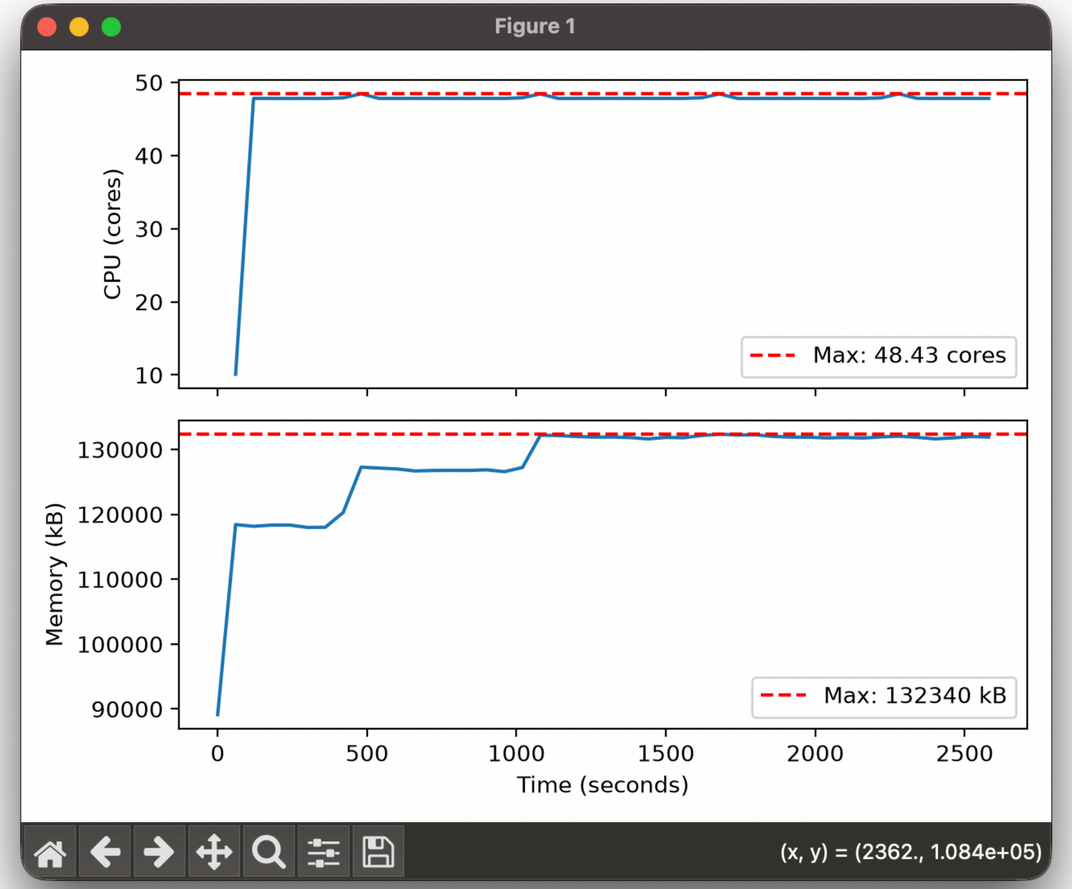
- Create a set of tools to:
  - Understand the data though *Exploratory Data Analysis* (EDA)
  - Forecast usage by making predictions using *Machine Learning* (ML)
  - Organize and schedule simulations on available clusters



Preliminary Results 1

NGA - *Lightweight Distributed Metric Service* (LDMS)

- Generation of sample data through *Large-scale Atomic Molecular Massively Parallel Simulator* (LAMMPS) simulations:
  - 223 simulations
  - Memory usage from 10 to 450 MB
  - CPU usage from 16 to 48 cores
- Visualizing simulation results



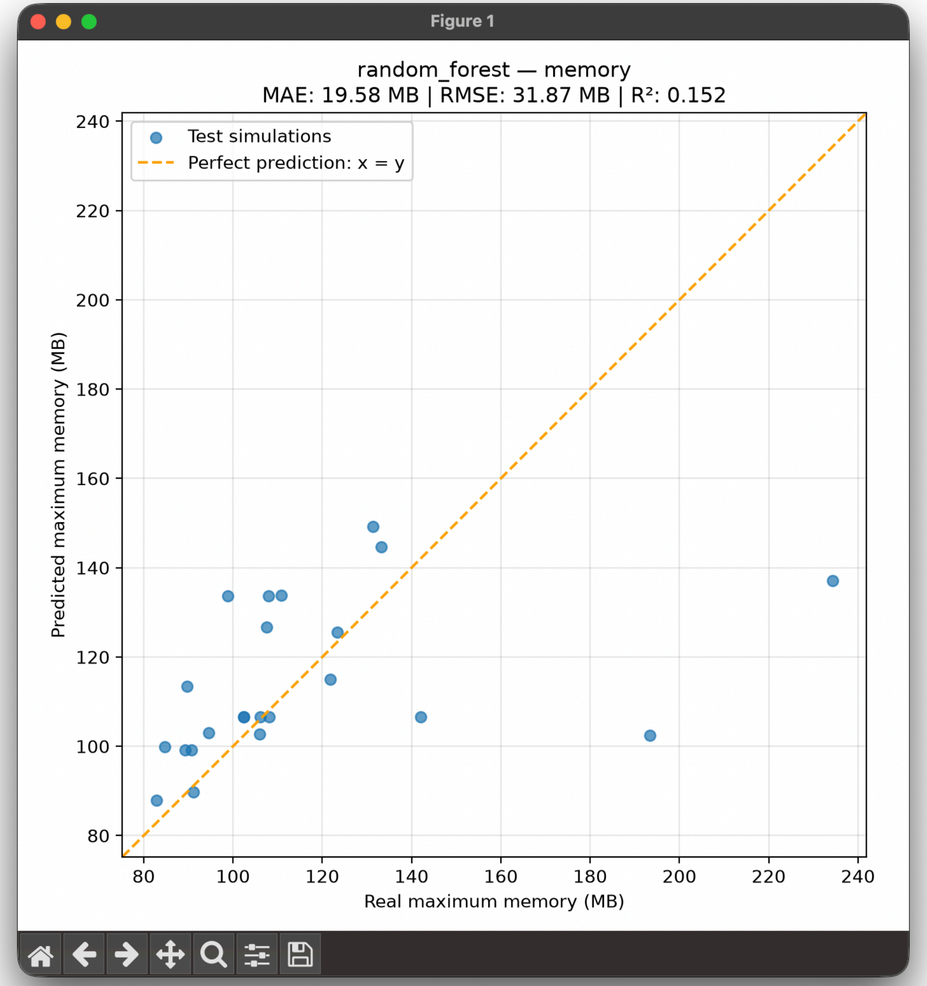
On the right: LAMMPS simulations of
131,072-atom charged LJ melt (+/-1 layered charges),
PPPM 1.0e-3 order 7, NVT and 80k steps

Preliminary Results 2

NGA - *Lightweight Distributed Metric Service* (LDMS)

- Training of various models: Ridge regression, *Multi-Layer Perceptron* (MLP), random forest
- Predicting peak CPU and memory usage
- Identify the best model

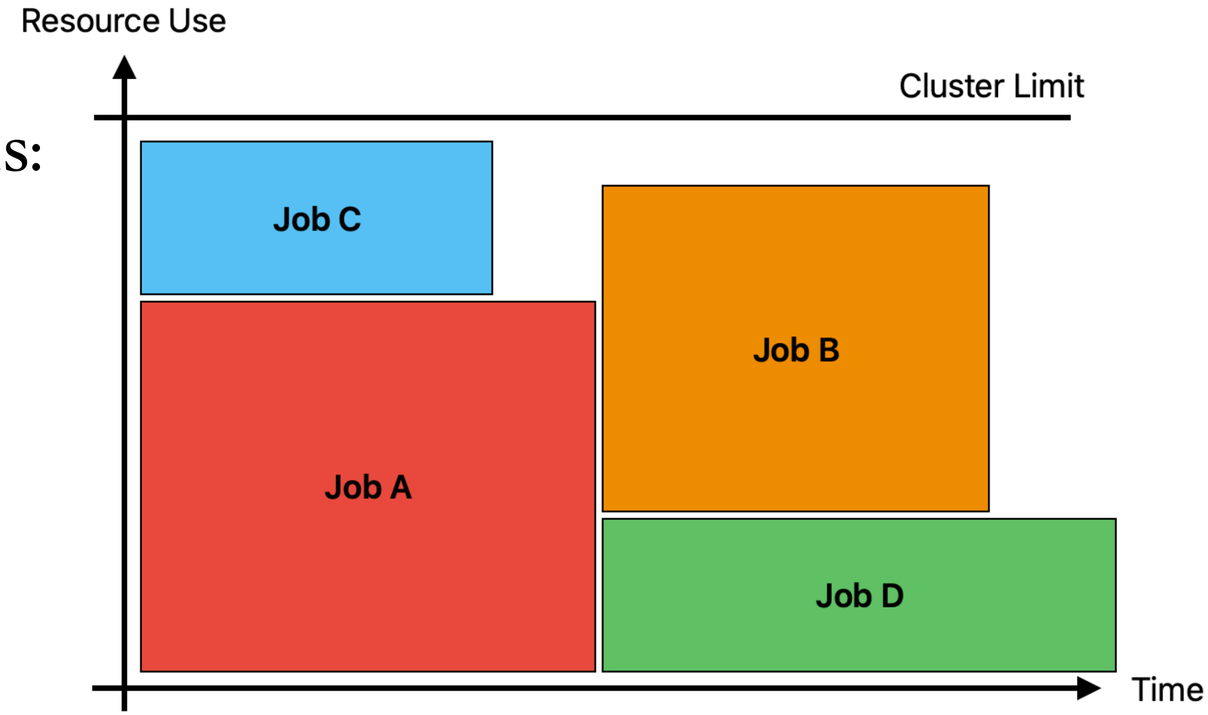
- We are also exploring a hash-based proximity search for the simulation's input parameters



Next Steps

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- Record more diverse simulation data
- Develop / use effective scheduling methods:
  - Heuristic
  - *Integer Linear Programming* (ILP)
- Expand to include other types of physical simulations (Gromacs, AMReX)



Questions?

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